Academic Year: 2025-26

Sap Id:60003240262

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Department of Information Technology

COURSE CODE: DJS23ILPC501 **DATE:23/9/2026**

COURSE NAME: Computer Networks **CLASS: T. Y. B. Tech**

Experiment No. 6

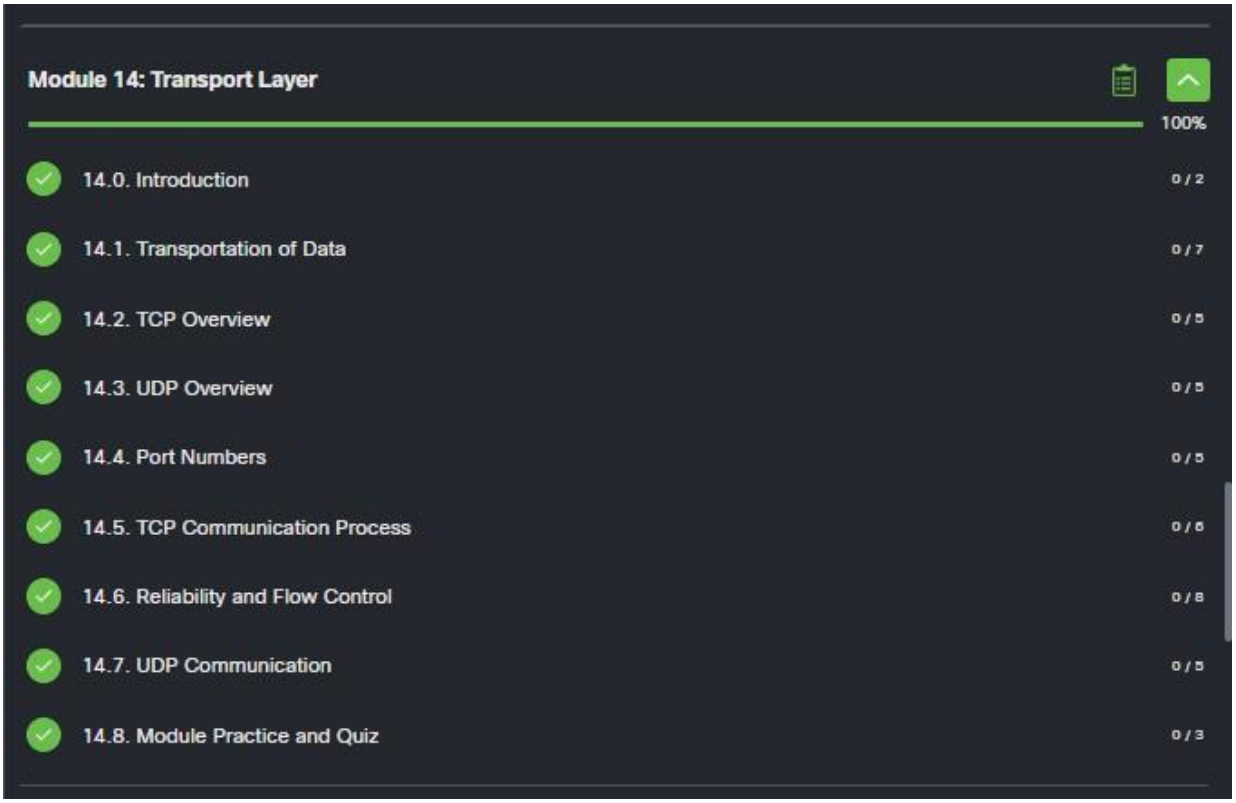
CO/LO: Analyse various concerns at different layers in reference models using suitable techniques.

Aim: **1.** To perform VLSM in packet tracer

2. Complete Module 14 of **CCNA**

Procedure:

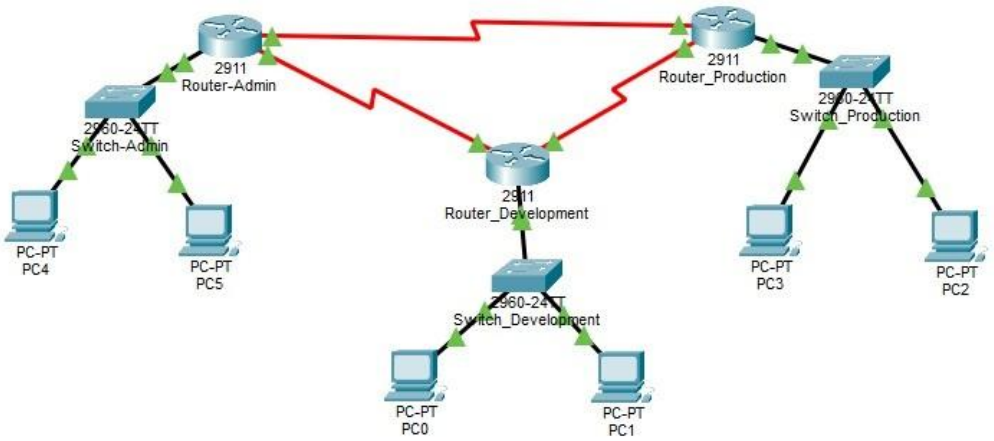
1. **Screenshot of Completion of Module 14**



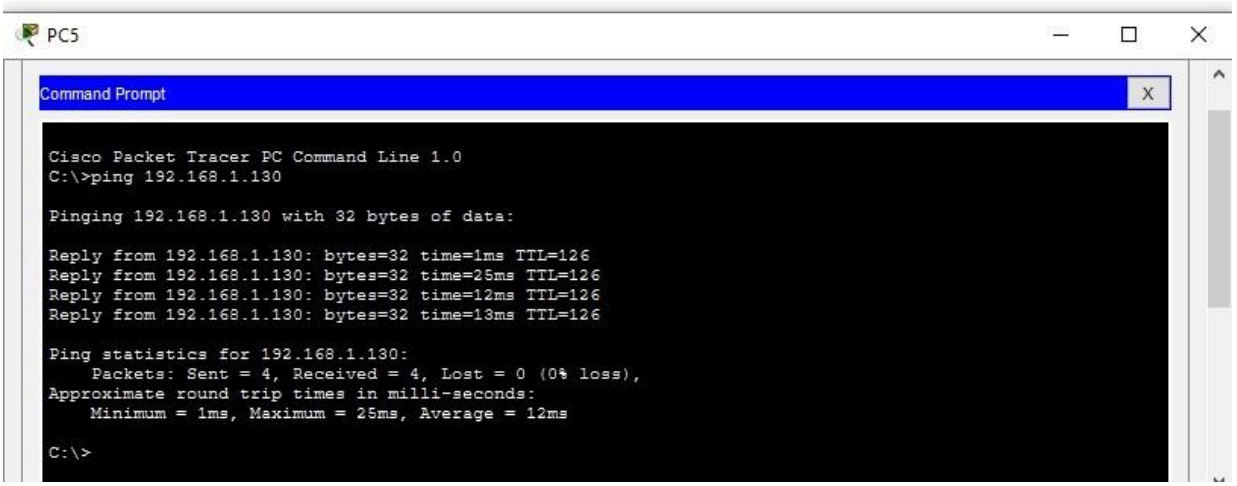
2. **Create six subnets three for departments (Three LAN's and three for WAN links) Subnet Designing: Sheet to be attached**

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Subnet	IP Address Requirement	CIDR	Network Address (Net ID)	First Host IP Address	Last Host IP Address	Broadcast IP Address	Subnet Mask
Subnet 1 – Development	74 hosts	/25	192.168.1.0	192.168.1.1	192.168.1.126	192.168.1.127	255.255.255.128
Subnet 2 – Production	52 hosts	/26	192.168.1.128	192.168.1.129	192.168.1.190	192.168.1.191	255.255.255.192
Subnet 3 – Administration	28 hosts	/27	192.168.1.192	192.168.1.193	192.168.1.222	192.168.1.223	255.255.255.224
Subnet 4 – WAN 1	2 hosts	/30	192.168.1.224	192.168.1.225	192.168.1.226	192.168.1.227	255.255.255.252
Subnet 5 – WAN 2	2 hosts	/30	192.168.1.228	192.168.1.229	192.168.1.230	192.168.1.231	255.255.255.252
Subnet 6 – WAN 3	2 hosts	/30	192.168.1.232	192.168.1.233	192.168.1.234	192.168.1.235	255.255.255.252



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Conclusions:

VLSM subnetting was successfully performed in Cisco Packet Tracer using the given IP address block 192.168.1.0/24. Six subnets were designed to meet the requirements of the three departments and three WAN links. Static routing was configured to enable communication between the departments, and connectivity was verified using the ping command. The experiment demonstrated efficient IP address allocation and inter-network communication.

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